

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, head, seek_time = 0;

    printf("Enter the number of disk requests: ");
    scanf("%d", &n);

    int request_queue[n];

    printf("Enter the disk request queue:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &request_queue[i]);
    }

    printf("Enter the initial position of the disk head: ");
    scanf("%d", &head);

    // Sort the request queue
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (request_queue[i] > request_queue[j])
            {
                int temp = request_queue[i];
                request_queue[i] = request_queue[j];
                request_queue[j] = temp;
            }
        }
    }

    // C-SCAN Scheduling
    printf("\nC-SCAN Disk Scheduling:\n");

    int start = 0;
    int end = n - 1;

    while (start <= end)
    {
        for (int i = start; i <= end; i++)
        {
            if (request_queue[i] >= head)
            {
                seek_time += abs(head - request_queue[i]);
                head = request_queue[i];
                start = i + 1;
            }
        }

        // Move head to beginning (circular jump)
        seek_time += abs(head - 0);
        head = 0;

        // Jump to highest request
        if (start <= end)
        {
            seek_time += abs(head - request_queue[end]);
            head = request_queue[end];
            end = end - 1;
        }
    }

    printf("Total Seek Time: %d\n", seek_time);
    printf("Average Seek Time: %.2f\n", (float)seek_time / n);

    return 0;
}
```

Output:

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Enter the number of disk requests: 3

Enter the disk request queue:

111

222

333

Enter the initial position of the disk head: 222

C-SCAN Disk Scheduling:

Total Seek Time: 444

Average Seek Time: 148.00

Process returned 0 (0x0) execution time : 9.782 s

Press any key to continue.